

HEWITT WATKINS

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EDUCATION

The University of Chicago, Chicago, IL

Expected Graduation June 2027 (B.S. & M.S.)

- *Master of Science in Computer Science*
- *Bachelor of Science in Computer Science* – GPA: 3.85/4.0
- Organizations & Affiliations: UChicago Human-Robot Interaction Lab, Kappa Theta Pi Technology Fraternity, Algo Group Algorithm Studies Club, Club Lacrosse (President), UChicago Football
- Relevant Coursework: Algorithms & Data Structures, Theory of Algorithms, Systems Programming, Database Systems, Machine Learning, Computer Vision, Unsupervised Learning, ML for Computer Systems

WORK EXPERIENCE

Capital One

June 2026 – August 2026

TIP Software Engineering Intern

Dallas, TX

- Built a real-time, event-driven document ingestion pipeline (Java, Kafka, AWS) processing ~1M documents/day, implementing intelligent filtering and fault-tolerant retries across distributed consumers
- Fine-tuned an internal model to extract key data from filtered financial documents, feeding structured output into downstream decisioning systems and eliminating manual review of financial applications
- Designed and implemented monitoring infrastructure, including automated failure alerting and diagnostic dashboard providing real-time visibility into pipeline health and enabling document throughput analysis
- Architected a multimodal fraud detection system combining computer-vision deep learning and LLM/OCR-based models to detect GenAI-generated content, image-level forgery, and manual document tampering

AlphaCore Wealth Advisory

June 2025 – August 2025

Software Engineering Intern

San Diego, CA

- Engineered internal Python tools to automate reporting across irregular, multi-source pipelines, improving consistency in time-sensitive financial datasets and reducing manual labor from hours to zero
- Jointly led firm-wide AI strategy and solution design, partnering with M&A, research, and advisory teams to build compliance-ready roadmap identifying data readiness, automation priorities, and resource inefficiencies

UChicago Human-Robot Interaction Lab

March 2024 – Present

Conversational AI Research Assistant

Chicago, IL

- Developed a novel multimodal VLM pipeline enabling unsupervised learning models to evaluate real-time robot-student conversations, providing structured analysis across 1,000 interactions in 3 Chicago schools
- Built a real-time social-norm-aware dialogue system using a local LLM to evaluate streaming speech-to-text partials, detecting rule violations as students speak; coordinated with non-local LLM for higher reasoning
- Engineered an end-to-end speech-to-speech pipeline with multithreading for continuous low-latency audio streaming, integrating Gemini and OpenAI APIs with prompt engineering and dynamic context windows

PUBLICATIONS

- “Fictional vs. Factual Robot Tutor Dialogue Can Shape Child Social-Emotional Learning,” *HRI '26*, 2026.

PROJECTS

Satellite Telemetry Anomaly Detection (Python, PyTorch)

- Built an unsupervised anomaly detection pipeline for 10M+ satellite telemetry records, implementing Variational Autoencoders (Conv1D, Hierarchical, Transformer), Gaussian Mixture Models, PCA, and K-Means to learn latent operational representations and detect real-time spacecraft system failures

Lempel-Ziv-Welch Compression and Decompression Tool (C)

- Built a high-performance, memory-efficient LZW compression tool in C with a scalable 2D lookup table, adaptive pruning, and dynamic bit-length encoding, achieving expected 50%+ compression rates
- Implemented custom bit-packing with buffered I/O, enabling low-latency encoding/decoding of large files

SQL Storage & Execution Engine (Rust)

- Created a slotted-page storage engine with heap file interface, buffer pool caching, and space reclamation
- Constructed and optimized query execution operators (nested loop join, hash join, group-by aggregation) in a volcano-style engine, balancing persistence, efficiency, and fault tolerance under database workloads

LANGUAGES

Python, Java, C/C++, Swift, Rust, SQL, Assembly, JavaScript, Bash, HTML, CSS, R, ROS

SOFTWARE DEVELOPMENT

Backend & Cloud: Kafka, Microservices, AWS (S3, Lambda, DynamoDB, Aurora, ECS), REST APIs, Spring Boot, Maven | Monitoring: CloudWatch, Splunk | Testing & Infra: JUnit, Postman, Docker | DevOps: Git, Jenkins

MACHINE LEARNING, MODEL DEVELOPMENT, & APPLIED AI

PyTorch, TensorFlow, Scikit-Learn, Deep Learning, Generative Models (VAEs, GANs), Diffusion Models, LLM Fine-Tuning, Local Model Deployment, Model Evaluation, Prompt/Context Engineering, Multimodal VLM